



YEAR 11 Mathematics Ext 1 Preliminary Course Assessment Task 4 2009

Time Allowed – 1Hour 30 Minutes

1. There are 5 questions, NOT of equal value.
2. Answer each question in the booklets provided.
3. Start each Question in a new booklet.
4. Show all necessary working
5. Use the ruled side of the page only.
6. Calculators may be used

Topic	Mark
1. Question 1 (Assorted topics)	/14
2. Question 2 (Polynomials)	/12
3. Question 3 (Counting Theory)	/12
4. Question 4 (Circle Geometry)	/8
5. Question 5 (Trigonometry)	/16

2009 – Extension 1 – Preliminary Higher School Certificate

Question 1 14 Marks

Marks

a) Differentiate $(x^2 + 1)\sqrt{x^2 + 1}$ **2**

b) Find the obtuse angle between the lines $y = 2x$ and $x + y - 3 = 0$ giving the answer correct to the nearest minute. **3**

c) $A(-2, 3)$ and $B(6, -1)$ are two points. Find the coordinates of the point P which divides the interval AB externally in the ratio 3 : 2. **3**

d) Solve and graph your solution on a number line: **3**

$$\frac{x^2 - 4}{x + 3} \leq x - 4$$

e) Simplify $\frac{2^{4n} \times 3^{2n}}{8^n \times 6^n} + 3^n + 3^n$ **3**

Question 2 (12 Marks) Begin a New Booklet**Marks**

- a) Find the remainder when $P(x) = x^3 - 3x^2 + 3x - 5$ is divided by $x - 2$ **2**

- b) The equation $x^3 + 3x^2 - 5x + 6 = 0$ has roots α , β and γ . Find the value of **2**

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

- c) (i) Use the Factor Theorem to show that $x^2 - 4$ is a factor of the polynomial **2**

$$P(x) = x^4 + 2x^3 - 3x^2 - 8x - 4$$

- (ii) Use long division to show that $P(x)$ has a **repeated** zero of $x = -1$ **2**

- (iii) Hence factorise $P(x)$ completely into linear factors. **1**

- (iv) Sketch the graph of $y = P(x)$ **2**

- (v) Hence or otherwise, solve $x^4 + 2x^3 - 3x^2 - 8x - 4 \geq 0$ **1**

Question 3 (12 Marks) Begin a New Booklet**Marks**

- a) (i) From a group of 7 girls and 6 boys, 3 girls and 2 boys are chosen. How many different groups of 5 are possible? **2**
- (ii) If the group of 5 stands in a line, how many arrangements will have the boys standing together? **2**
- b) (i) Bob chooses six numbers from the numbers 1 to 40 inclusive. In how many ways can he do this? **1**
- (ii) A machine then chooses six numbers from the numbers 1 to 40 inclusive. In how many ways can the machine's numbers all be different from any of Bob's numbers? **1**
- c) The letters from the word KURRARONG are cut out, and placed in a hat.
- (i) All of the nine letters are drawn out one at a time and placed in a line, to form a "word". How many different "words" are possible? **1**
- The letters are then replaced in the hat and mixed well.
- (ii) Four letters are drawn out at random and placed in a line, to form a "word". How many different four letter "words" can be made in this way? **3**
- d) Show that $\frac{1}{(n-1)!} - \frac{1}{n!} + \frac{1}{(n+1)!} = \frac{n^2}{(n+1)!}$ **2**

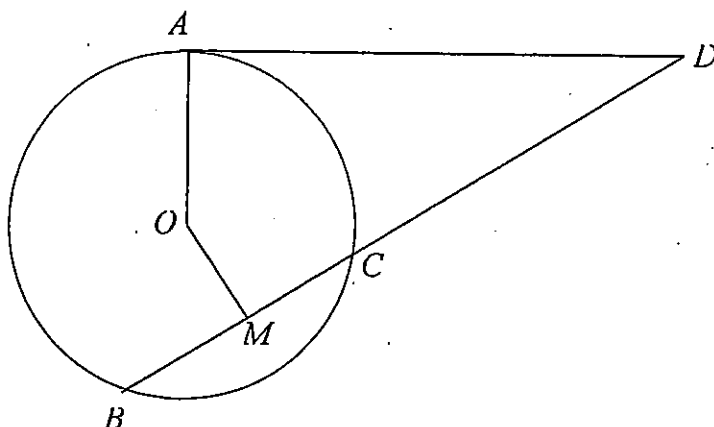
Question 4 (8 Marks) Begin a New Booklet

Marks

- a) A, B and C are points on a circle with centre O. The tangent to the circle at A meets BC produced at D. M is the midpoint of BC.

(i) Copy the diagram and show that AOMD is a cyclic quadrilateral. **3**

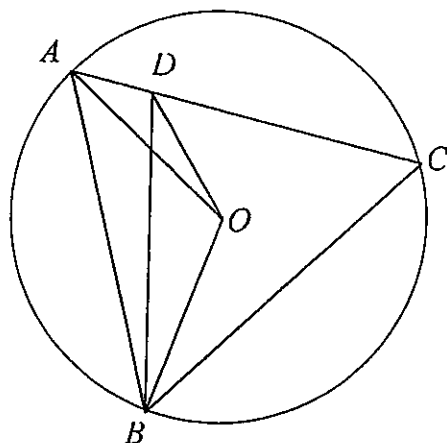
(ii) Give a reason why OD is a diameter of the circle through A, O, M and D. **1**



- b) ABC is a triangle inscribed in a circle with centre O. D is a point on CA such that ABOD is a cyclic quadrilateral.

(i) Copy the diagram. Give a reason why $\angle BOA = \angle BDA$. **1**

(ii) Hence show that $BD = CD$. **3**

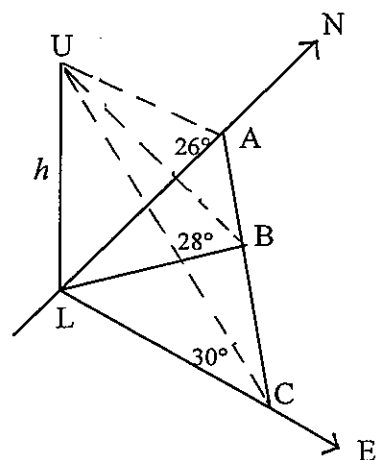


Question 5 (16 Marks) Begin a New Booklet

Marks

- a) Express $\tan(90^\circ + \theta)$ in terms of $\tan \theta$ **1**
- b) Solve the equation $\cos 2x + \sin x = 0$ for $0^\circ \leq x \leq 360^\circ$ **4**
- c) Prove that $\frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta$ **3**
- d) Use the substitution $t = \tan \frac{x}{2}$ to show that $\frac{\sin x}{1 - \cos x} = \cot \frac{x}{2}$ **2**

- e) Uluru is a large rock h metres high, on flat ground in Central Australia. Three tourists A, B, and C are observing Uluru from the ground. A is due north of Uluru, C is due east of Uluru, and B is on the line-of-sight from A to C and between them. The angles of elevation to the summit of Uluru from A, B, and C are 26° , 28° , and 30° , respectively.



- (i) Show that the distance of A from the base L of Uluru is $h \tan 64^\circ$ **1**
- (ii) Give similar expressions for each of the distances of B and C from the base L. **1**
- (iii) Show that $\angle LAC$ is $40^\circ 11'$ to the nearest minute. **1**
- (iv) Hence, determine the bearing of B from the base of Uluru (to nearest degree). **3**

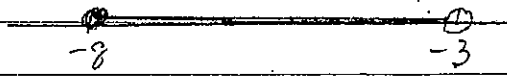
Solutions to 2009 Preliminary HSC Ext 1

1 a) $y = (x^2 + 1)^{3/2}$
 $y' = \frac{3}{2} (x^2 + 1)^{1/2} \cdot 2x$
 $y' = 3x \sqrt{x^2 + 1}$

b) $m_1 = 2, m_2 = -1$ ($y = -x + 3$)
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \frac{2 + 1}{1 - 2} = \left| \frac{3}{-1} \right| = 3$
 $\tan \theta = 3 \quad \therefore \theta = 180 - 71^\circ 34'$
 $\theta = 108^\circ 26'$

c) $A(-2, 3) \quad B(6, -1) \quad k:l = 3:-2$
 $x = \frac{3 \times 6 + (-2) \times (-2)}{3 - 2} \quad y = \frac{3 \times (-1) + (-2) \times 3}{3 - 2}$
 $x = 22, y = -9 \quad \therefore P \text{ is } (22, -9)$

d) $\frac{x^2 - 4}{x + 3} < x - 4$
 For $x > -3$ solve $x \neq -3$ For $x < -3$ solve
 $x^2 - 4 \leq (x - 4)(x + 3)$ $x^2 - 4 \geq (x - 4)(x + 3)$
 $x^2 - 4 \leq x^2 - x - 12$ $x^2 - 4 \geq x^2 - x - 12$
 $x \leq -8$ $x \geq -8$
 No Solution


 $-8 \leq x \leq -3$

e) $\frac{2^{4n} \times 3^{2n}}{8^n \times 6^n} + 3^n + 3^n$
 $\frac{2^{4n} \times 3^{2n}}{2^{3n} \times 2^n \times 3^n} + 3^n + 3^n$
 $3^n + 3^n + 3^n$
 $= 3 \times 3^n$
 $= 3^{n+1}$

Question 2

a) $p(x) = x^3 - 3x^2 + 3x - 5$

$$p(2) = 8 - 12 + 6 - 5 = \underline{\underline{-3}}$$

b) $x^3 + 3x^2 - 5x + 6 = 0$

$$\Sigma \alpha = -3$$

$$\Sigma \alpha\beta = -5$$

$$\Sigma \alpha\beta\gamma = -6$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{-5}{-6} = \underline{\underline{\frac{5}{6}}}$$

c) $x^2 - 4 = (x-2)(x+2)$

$$p(2) = 16 + 16 - 12 - 16 - 4 = 0 \therefore (x-2) \text{ is factor}$$

$$p(-2) = 16 - 16 - 12 + 16 - 4 = 0 \therefore (x+2) \text{ is factor}$$

$\therefore (x-2) + (x+2)$ are factors i.e. $(x^2 - 4)$

ii) Repeated zero of $x = -1 \therefore$ repeated factor of $(x+1)^2$ i.e. $(x+1)^2$

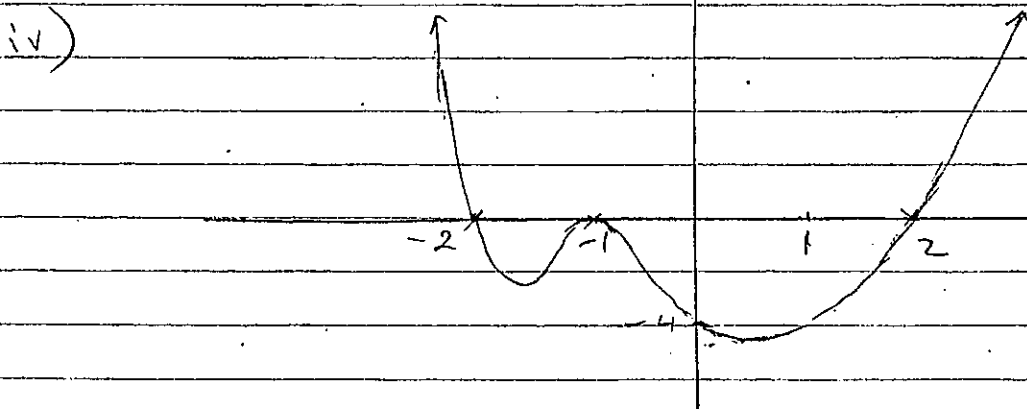
$$\begin{array}{r} x^2 - 4 \\ x^2 + 2x + 1 \end{array} \quad \begin{array}{r} x^4 + 2x^3 - 3x^2 - 8x - 4 \\ x^4 + 2x^3 + x^2 \end{array}$$

$$-4x^2 - 8x - 4$$

$$-4x^2 - 8x - 4$$

Remainder is zero $\therefore (x+1)^2$ is a factor
+ $x = -1$ is a repeated zero.

iii) $p(x) = (x-2)(x+2)(x+1)^2$



v) $y \geq 0$ for $x \leq -2$, $x = -1$, $x \geq 2$

$\therefore$ Solution is $x \leq -2$ or $x = -1$ or $x \geq 2$

Question 3

a) i) ${}^7C_3 \times {}^6C_2 = 525$

ii) Put the boys together 2 ways
Arrange 4 things in a line 4!
 $\therefore 2 \times 4! = 48$

b) i) ${}^{40}C_6$

ii) ${}^{34}C_6$

c) ii) 3 R's 1 other: Choose 6C_1 & arrange 4 = ${}^6C_1 \times 4$

2 R's 2 other: choose 6C_2 & arrange $\frac{4!}{2!} = {}^6C_2 \times \frac{4!}{2!}$

4 different letters: choose & arrange 7P_4

$$= 24 + 180 + 840 = 1044 \text{ words}$$

d) $\frac{1}{(n-1)!} - \frac{1}{n!} + \frac{1}{(n+1)!}$

$$= \frac{n(n+1) - (n+1) + 1}{(n+1)!}$$

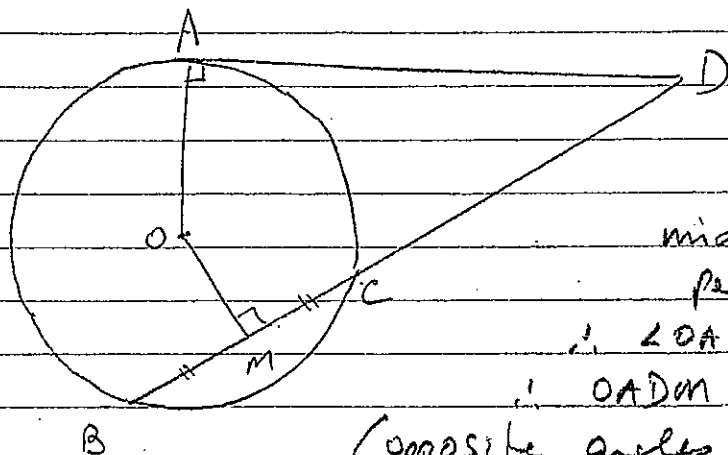
$$= \frac{n^2 + n - n - 1 + 1}{(n+1)!}$$

$$= \frac{n^2}{(n+1)!}$$

c) i) $\frac{9!}{3!}$

Question 4

a)



$\angle OAD = 90^\circ$
(Angle between tangent & radius)

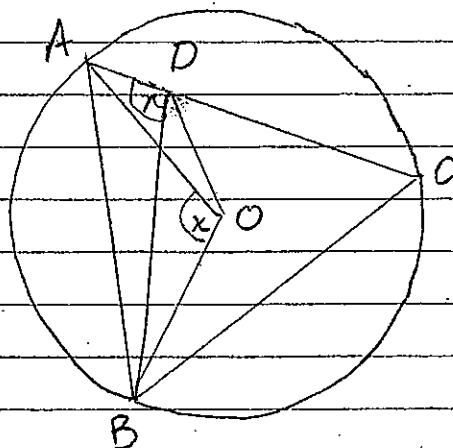
$$\angle OMC = 90^\circ$$

(line from centre to mid point of chord is perpendicular to chord)

$$\therefore \angle OAD + \angle OMC = 180^\circ$$

$\therefore$ OADM is a cyclic quadrilateral
(opposite angles are supplementary)

ii) OD is diameter because the angle in a semi-circle is a right angle.



i) $\angle BOA = \angle BDA = x$ Angles at circumference standing on minor arc AB of circle ABO D

ii) $\angle BDC = 180 - x$ (straight $\angle$)
 $\angle ACB = \frac{x}{2}$ Angle at circumference is half angle at centre standing on arc AB of circle ABC

$\therefore \angle DBC = x - \frac{x}{2}$ (exterior angle equals sum of interior opposite angles)

$$\therefore \angle DBC = \frac{x}{2}$$

$$\therefore \angle DBC = \angle DCB \quad (\angle ACB)$$

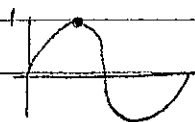
$\therefore BD = DC$ ($\triangle DBC$ is isosceles, sides opposite equal angles are equal)

Question 5

$$\begin{aligned}
 a) \quad \tan(90+\theta) &= \frac{\sin(90+\theta)}{\cos(90+\theta)} = \frac{\sin 90 \cos \theta + \cos 90 \sin \theta}{\cos 90 \cos \theta - \sin 90 \sin \theta} \\
 &= \frac{\cos \theta + 0}{0 - \sin \theta} = -\cot \theta \\
 &= \frac{-1}{\tan \theta} \quad (\text{other methods})
 \end{aligned}$$

$$\begin{aligned}
 b) \quad \cos 2x + \sin x &= 0 \\
 1 - 2\sin^2 x + \sin x &= 0 \\
 2\sin^2 x - \sin x - 1 &= 0 \\
 (2\sin x + 1)(\sin x - 1) &= 0 \\
 \sin x = -\frac{1}{2} \quad \text{or} \quad \sin x = 1 \\
 x = 210^\circ \quad \text{or} \quad 330^\circ \quad \text{or} \quad 90^\circ
 \end{aligned}$$

$$\begin{array}{r}
 2\sin x + 1 \\
 \times \\
 \sin x - 1 \\
 \hline
 \end{array}$$



$$\begin{aligned}
 c) \quad \frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} &= \frac{\frac{2\tan \theta}{1-\tan^2 \theta} - \tan \theta}{\frac{2\tan \theta}{1-\tan^2 \theta} + \frac{1}{\tan \theta}} \\
 &= \frac{2\tan \theta - \tan \theta + \tan^3 \theta}{(1-\tan^2 \theta)} \times \frac{(1-\tan^2 \theta) \tan \theta}{2\tan^2 \theta + 1 - \tan^2 \theta} \\
 &= \frac{\tan \theta (1+\tan^2 \theta) \times \tan \theta}{\tan^2 \theta + 1} \\
 &= \tan^2 \theta
 \end{aligned}$$

$$\begin{aligned}
 d) \quad \frac{\sin x}{1-\cos x} &= \frac{2t}{1+t^2} \div \left(1 - \frac{1-t^2}{1+t^2}\right) \\
 &= \frac{2t}{1+t^2} \div \left(\frac{1+t^2 - 1+t^2}{1+t^2}\right) \\
 &= \frac{2t}{1+t^2} \times \frac{1+t^2}{2t^2} \\
 &= \frac{1}{t} \\
 &= \csc \frac{x}{2}
 \end{aligned}$$

5 e)

i) $\tan 26^\circ = \frac{h}{AL}$ in $\triangle ALU$

$$\therefore AL = \frac{h}{\tan 26^\circ} = \frac{h}{\cot 64^\circ}$$

$$AL = h \tan 64^\circ$$

ii) $BL = h \tan 62^\circ$

$$CL = h \tan 60^\circ$$

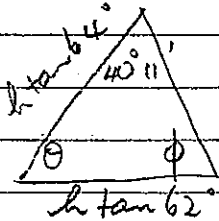
iii) In $\triangle ALC$

$$\tan \angle LAC = \frac{LC}{LA} = \frac{h \tan 60^\circ}{h \tan 64^\circ}$$

$$\angle LAC = \tan^{-1} \left(\frac{\tan 60^\circ}{\tan 64^\circ} \right) = \tan^{-1} 0.8447776$$

$$\angle LAC = 40^\circ 11' \text{ to nearest minute}$$

iv) In $\triangle ALB$



$$\frac{\sin \phi}{h \tan 64^\circ} = \frac{\sin 40^\circ 11'}{h \tan 62^\circ}$$

$$\sin \phi = \frac{\tan 64^\circ \sin 40^\circ 11'}{\tan 62^\circ}$$

$$\sin \phi = 0.703516$$

$$\angle \phi = 44^\circ 43' \text{ or } 180 - 44^\circ 43' = 135^\circ 17'$$

$$\therefore \angle \theta = 180 - 44^\circ 43' - 40^\circ 11' \text{ or } 180 - 135^\circ 17' - 40^\circ 11'$$

$$\angle \theta = 95^\circ 6'$$

$$\begin{aligned} &4.533^\circ \\ &4^\circ 32' \end{aligned}$$

But B is between A + C $\therefore$ obtuse angle is not possible

$\therefore$ Bearing is $4^\circ 32'$ or $005^\circ T$
or $N 5^\circ E$ to nearest degree